European Polymer Journal 226 (2025) 113710

Contents lists available at ScienceDirect

European Polymer Journal

journal homepage: www.elsevier.com/locate/europolj

Concentration-independent molecular weight determination of polymers via diffusion NMR: A universal approach across solvents

Francisco Manuel Arrabal-Campos [b,*], Marta González-Lázaro [a], Juana M. Pérez [a], Juan A. Martínez Lao [b], Ignacio Fernández [a,*]

[a] Ctra, Sacramento s/n, Department of Chemistry and Physics, Research Center CIAIMBITAL, University of Almería, 04120 Almería, Spain
[b] Ctra, Sacramento s/n, Department of Engineering, Research Center CIMEDES, University of Almería, 04120 Almería, Spain

ARTICLE INFO

Keywords:
Pulse Field Gradient Diffusion
Diffusion ordered spectroscopy
Polymers
NMR
Universal Calibration Curve

ABSTRACT

Diffusion NMR is a widely used technique to measure diffusion coefficients, which provides insights into the size and mobility of molecules in solution. For polymer and other systems, these diffusion coefficients are often linked to molecular weight through empirical scaling relationships, allowing the molecular weight to be estimated indirectly. However, factors such as solvent type and polymer concentration can have a significant impact on the precision of molecular weight predictions derived from these coefficients. Typically, diffusion NMR measurements are performed at very low concentrations to minimize distortions in diffusion coefficients, which can affect molecular weight estimation. However, this approach poses challenges, such as inaccuracies due to the need to very small quantities of polymer. Additionally, achieving reliable results often requires numerous scans and long recovery periods between scans to account for T1 relaxation times, leading to extended experimental durations. To avoid these drawbacks, we propose a new iterative method that incorporate the relationship $D\eta|_c = D\eta|_{1/\infty} e^{-\kappa C^{\psi}}$ where the Kappa value depends on the molecular weight affording a new scaling law $D\eta|_c = ae^{-(mMw+n)C^{\psi}} M_w^{-b}$. This approach improves the accuracy of molecular weight estimates described so far and makes it independent to changes in the type of solvent and more importantly in the concentration of the polymer validate from 1.5 mg/mL until 150 mg/mL. This method represents a step forward in the precise characterization of polymers using diffusion NMR.

1. Introduction

Pulse Field Gradient Spin Echo (PFGSE) Nuclear Magnetic Resonance (NMR) experiments are a type of NMR methodology used for quantitative analysis of molecular diffusion. This technique powers the ability to spatially encode the diffusion of molecules through the application of a pulse field gradient. [1] PGSE NMR is often combined with other NMR methods to provide detailed information about the structure and dynamics of molecules in a sample. It is widely used in chemistry, biology, and materials science [2,3] because it can provide valuable insights into the size, shape, molecular weight, and interactions of molecules with other substances. [4,5] In the last years, there has been particular interest in using this methodology to determine the molecular weight of polymers. [6].

The measurable physical property is the diffusion coefficient, which is closely linked to the molecular weight through its corresponding calibration curve. [7] Several research groups have made significant contributions to the development of weight-average molecular weight (Mw) prediction methods using diffusion NMR through the use of DOSY (Diffusion Ordered Spectroscopy) methodologies. These include the teams led by Williard,[8] Johnson,[9] Grubbs, [10] and Hiller [11] among others. [12] The empirical correlation between the diffusion coefficient and molecular weight is effectively explained by the Flory's regime scaling law,[13] which stands as the predominant model. This law is expressed as follows,

$$D = \alpha M_w^{-\beta} \tag{1}$$

where α and β are the scaling parameters [14] that depend on the polymeric system used and its conformation in each solvent. D and Mw are the diffusion coefficient and weight-average molecular weight, respectively. Delsuc and colleagues have conducted extensive validation of the parameter β across a variety of polymeric systems. Furthermore,

* Corresponding authors.
E-mail address: ifernan@ual.es (I. Fernández).

https://doi.org/10.1016/j.eurpolymj.2024.113710
Received 12 October 2024; Received in revised form 29 December 2024; Accepted 31 December 2024
Available online 10 January 2025
0014-3057/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

they established a relationship between this parameter, denoted as β, and the geometry of the polymers by defining the fractal dimension as the inverse of β. [15] The values of β vary typically from 3/4 in the case of non-spherical systems or conformation in solution to 1/3 for polymers with spherical shape. [16] In this context, the work by Pollack demonstrated that the diffusion coefficient (D) exhibits a power-law dependence on viscosity, [17] which significantly deviates from the predictions of the Stokes–Einstein equation in the diluted concentration regime. Pollack attributed this deviation to the interaction of the diffusing particle with the viscous medium, rather than to the shape or size of the particle itself. This finding underscores the importance of considering medium-specific interactions in diffusion studies, particularly in systems with varying viscosities. In addition, Pollack later developed a new scaling law relating diffusion coefficient, viscosity and temperature, while retaining the same principles of his earlier work. [18] Based on this approach, we reconsidered eq (1) and obtained equation (2) as a linearized relation that is independent of viscosity. [19]

$$D\eta = \alpha M_w^{-\gamma} \tag{2}$$

where α is a scaling parameter and γ is equal to 1/3β, where β describes the interaction between solvent and polymer, explaining the solvent structuring driven by the functional groups of the polymer. [19] In equation (2), the viscosity is denoted with η, and D and Mw are the diffusion coefficient and molecular weight, respectively. This relationship enabled the construction of a universal calibration curve that is not dependent on the chosen deuterated solvent. The primary drawback was the necessity of preparing polymer solutions under highly diluted conditions (usually around 1 mg/mL), but in favor is that constant instrument recalibration is not needed, rendering diffusion NMR considerably more accessible [20] as a tool for determining molecular weight compared to other methods such as SEC, the current standard practice. However, SEC is widely regarded for its ease of use, lower initial instrument cost, and minimal parameter optimization requirements, making it an accessible choice for many laboratories.

Recent works have significantly enhanced these methodologies. For instance, a universal calibration approach was developed utilizing the Mark-Houwink equation to provide structure- and solvent-independent molar mass calibration for DOSY. This method aligns closely with SEC data, making it a valuable tool for determining molar masses and calculating new Mark-Houwink parameters using DOSY. [11] Ruzicka et al. have underscored the importance of careful preparation required for generating calibration curves. In their study, they analyzed variables such as pulse sequence, polymer dispersity, and sample concentration. [21] Importantly, their findings align with the majority of studies indicating that reliable data was consistently obtained at sample concentrations of 0.5 mg/mL. Recently, the introduction of MaDDOSY (Mass Determination Diffusion Ordered Spectroscopy) with an 80 MHz benchtop NMR has initiated the implementation of known molecular weight estimation techniques with diffusion sequences on benchtop spectrometers. In the contribution by O. Tooley et al., they describe a method that is independent of solvent and polymer type, overcoming the limitations of high-field NMR instruments and making molecular weight determination down to 10 mg/mL more practical and accessible. [22] Moreover, Silva et al. have performed temperature dependence investigations of DOSY experiments, proving capable of extending the molar mass range and temperature window for calibration, achieving good agreement with the Stokes–Einstein equation. [23] As previously mentioned, all these studies employed very diluted conditions, highlighting the need to develop methods that do not require low concentrations. Typically, the diffusion coefficient of a polymer exhibits a direct dependence on its concentration, C. Based on the models proposed by Edwards [24] and de Gennes, [25] polymeric solutions are classified into three distinct regimes: dilute, semi-dilute, and concentrated. The self-diffusion behavior of polymers varies significantly across these regimes, reflecting the differing characteristics at each concentration level.

In dilute solutions, the diffusion coefficient (D) is influenced by the hydrodynamic radius (Rh), which represents the effective size of the polymer as it moves through the solvent. The radius of gyration (Rg), a measure of the polymer's spatial distribution, is proportional to Rh, with the proportionality constant depending on the polymer conformation degree of flexibility, and solvent interactions. Schaeffer et al. [26] introduced the concept of a marginal solvent to describe polymer dynamics under near- θ conditions, where polymer–solvent interactions balance those based on polymer–polymer. This framework also encompasses the mean-field approximation and diffusion within a reptation model, where transverse diffusion along a confining tube is significantly slower than the polymer's overall diffusion. This relationship is captured in Eq. (3), [27] in where Mw is the polymer weight-average molecular weight, C is the concentration and X is a proportionality constant wich is assumed to be independent of Mw or solvent.

$$R_h \sim XR_g \sim D M_w^{-2} C^{-2.5} \tag{3}$$

Some years later, light scattering experiments of optical probe particles of known hydrodynamic radii (Rh) in polymeric solutions of PEO (polyethylene oxide), PAA (polyacrylic acid) and BSA (bovine serum albumin) allowed the introduction of a new scaling law (Eq. (4). [28]

$$D = D_0 e^{-aC^{\nu} M_w^{\gamma} R_h^{\delta}} \tag{4}$$

where D_0 is the diffusion coefficient of a single molecule in infinitely dilute solution, where there are no interactions with other molecules. C and M_w denote the polymer concentration and weight-average molecular weight, respectively, while Rh is the hydrodynamic radius of the molecule used as a probe. The parameters ν, γ, and δ are scaling coefficients describing how the diffusion coefficient (D) in the non-infinity dilute state depends on these variables. Experimentally values of v, γ and δ are 0.6–1.0, 0.8 ± 0.1, and $\delta = 0$, respectively, that are all of them inconsistent with theoretical predictions. In the report by Phillies [29] the diffusive properties of several polymers as a function of concentration in different solvent systems were determined, in where logarithmic plots of D vs concentration showed smooth curves of progressively increasing curvature, that were only fitted with equation (5),

$$D = D_0 e^{-\kappa C^{\nu}} \tag{5}$$

where D_0 is the diffusion coefficient of single molecule, and the scaling coefficient κ is now dependent on the molecular weight. The experimental results indicated that the solutions of both polymers and globular proteins [30] follow a scaling law similar to that expressed in Eq. (5), leading to the conclusion that all macromolecules adhere to a universal scaling law. In line with this, Ullmann and colleagues [31] explored the behavior of polystyrene latex spheres in a polyethylene oxide and water mixture, reinforcing findings previously established by Phillies and others. [26] They observed that the relationship between viscosity and the diffusion coefficient deviated from the traditional Stokes-Einstein behavior and this relation depended on the molecular weight of the polymer, its concentration, and hydrodynamic radii, suggesting that viscosity also exhibited a concentration-dependent scaling law given in Eq. (6),

$$\eta = \eta_0 e^{-\kappa C^{\nu}} \tag{6}$$

where η is the viscosity of the polymer–solvent mixture, η_0 is the viscosity of the solvent in the absence of polymer, and both α and ν are the coefficients of the scaling law. This relationship has been observed in various polymer–solvent systems, including water and polyethylene oxide (PEO) mixtures, polyacrylic acid (PAA), and bovine serum albumin (BSA), often with added salts or buffers to control ionic strength. The polymer concentration ranges studied span from 0 to 350 mg/mL, encompassing dilute to highly viscous regimes, depending on the

molecular weight and type of polymer. The remarkable parallel of Eqs. (5) and (6) sets the foundations for the development of a new universal scaling law, which is able to cover a wide range of solvents and concentrations. This merger of relationships is given in Eq. (7) and allows for a more coherent interpretation of the polymer–solvent interactions at different concentration levels, always assuming high dilution as mentioned previously.

$$D\eta|_c = D\eta|_0 e^{-\kappa C^{\nu}} \tag{7}$$

In this equation $D\eta|_c$ denotes the product between the diffusion coefficient and the viscosity at a specific concentration, while $D\eta|_0$ represents the same product under dilute conditions. The Eq. (7) can also be transformed into a logarithmic equation (Eq. (8)

$$\ln(D\eta|_c) = -\kappa C^{\nu} + \ln(D\eta|_0) \tag{8}$$

Interestingly, if we consider a situation in where the concentration is zero, then the term $D\eta|_0$ can be changed by $D\eta|_{1/\infty}$, and the regime would be named infinite dilution. Eq. (8) takes now the form of Eq. (9).

$$\ln(D\eta|_c) = -\kappa C^{\nu} + \ln(D\eta|_{1/\infty}) \tag{9}$$

In the field of polymer science, it has been established that the scaling coefficient ν is approximately 1 for polymers [32,36]. Furthermore, the product of diffusion coefficient and viscosity at the limit of infinite dilution, $D\eta|_{1/\infty}$ exhibits the same scaling law than in Eq. (2), but in this case the coefficients α and γ are replaced by a and b, respectively, affording Eq. (10).

$$D\eta|_{1/\infty} = aM_w^{-b} \tag{10}$$

The Eq. (10) can also be transformed into a logarithmic equation (Eq. (11)

$$\ln(D\eta|_{1/\infty}) = -\mathrm{b}\log(\mathrm{M_w}) + \mathrm{a} \tag{11}$$

This new regime of real infinite dilution gives rise to a conceptual leap where the consequences in polymer physics are yet to be investigated. The proportionality coefficient κ (Eq. (5), as already mentioned, depends on the molecular weight, and follows a linear function as given in Eq. (12), being e and d two new scaling coefficients.

$$\kappa = \mathrm{m}\,M_w + \mathrm{n} \tag{12}$$

The conjugation of Eq. (7) and Eq. (12) gives a new scaling law that incorporates the molecular weight dependence with coefficient κ, yielding Eq. (13).

$$D\eta|_c = ae^{-(mMw+n)C^{\nu}}\,M_w^{-\mathrm{b}} \tag{13}$$

2. Materials and methods

2.1. Materials

Polystyrene samples (PS1 to PS8) were purchased from Polymer Standards Service (Mainz, Germany) as part of Agilent. Polypropylene glycol (PPG1, PPG2, PPG3 and PPG4) were purchased from American Polymer Standards Corporation (Ohio, USA). Table 1 summarizes the Mw, Mn and PDI values for all of them. Tetrahydrofuran-d_8 (THF-d_8), toluene-d_8 (Tol-d_8) benzene-d_6 (C_6D_6) and chloroform-d_1 ($CDCl_3$) were purchased from Eurisotop (Saint-Aubin, France). THF-d_8 and Tol-d_8 were dried over K/benzophenone, reflux and vacuum transferred onto 3 Å molecular sieves prior to use. C_6D_6 and $CDCl_3$ were dried over CaH_2 and vacuum transferred onto 3 Å molecular sieves prior to use. All other reagents and solvents were of commercial quality and were used without further purification. The polymers used in this study were deliberately selected for their narrow dispersity (polydispersity index as close as 1.0) to ensure accuracy for providing a clear validation of the approach. These monodisperse samples allowed us to isolate the effects of molecular weight and concentration from dispersity-related variability. Details on the dispersity values for all samples are provided in Table 1.

Table 1
Mw, Mn and PDI values for Polystyrene PS1-PS8 and Polypropylene glycol PPG1-PPG4 samples[a].

PS	Mw (Da)	Mn (Da)	PDI	PPG	Mw (Da)	Mn (Da)	PDI
PS1	370	370	1.00	PPG1	460	410	1.12
PS2	1070	968	1.11	PPG2	1150	990	1.16
PS3	34000	33200	1.02	PPG3	3250	3050	1.07
PS4	58900	53800	1.09	PPG4	5300	4800	1.10
PS5	93400	55100	1.70				
PS6	133000	124000	1.07				
PS7	335000	322000	1.04				
PS8	543000	505000	1.08				

[a] Polystyrene (PS) polymers used in this study were obtained from Agilent Technologies and conform to the US OSHA Hazard Communication Standard 29CFR1910.1200. Similarly, the polypropylene glycol (PPG) polymers used were obtained from American Polymer Standards Corporation and also conform with the same Hazard Communication Standard. Safety Data Sheets (SDS) for both materials have been included in the supplementary information.

2.2. NMR

All measurements were performed on a Bruker Avance III 500 spectrometer equipped with a microprocessor-controlled gradient unit and a third radiofrequency channel using an indirect 5 mm TBI ^{1}H/^{31}P/BB triple probe with an actively shielded Z-gradient coil. The spectral reference used was internal tetramethylsilane for ^{1}H. The shape of the gradient pulse was smoothed rectangular shape (SMSQ), and its strength varied automatically during the experiments. The calibration of the gradients on each spectrometer was carried out via a diffusion measurement of HDO in D_2O at 294 K. The D value of each sample can then be calculated according to the following equation, where $D_{HDO} = 1.9\ 10^{-9}\ m^2/s$. [33]

$$D = \frac{m_t}{m_{HDO}} D_{HDO}$$

The m_t parameter is calculated for each measurement from the observed slope, slope, with the help of the next equation, which considers the settings for the diffusion delay Δ, the gradient length δ, as well as the nature of the observed nucleus γ, with respect to the calibration with HDO.

$$m_t = \frac{slope}{\left(\frac{\gamma}{\gamma_{HDO}}\right)^2 \left(\frac{\delta}{\delta_{HDO}}\right)^2 \frac{\left(\Delta - \frac{\delta}{4}\right)}{\left(\Delta - \frac{\delta}{4}\right)_{HDO}}}$$

For PS (PS1 to PS8) and PPG (PPG1 to PPG4) standard samples each NMR tube contained the mg described in Tables S1-S12. The Δ values were 50–100 ms for P1-P6 and 100–150 ms for P7 and P8. The δ values were 2.75 (PS1), 3.2 (PS2), 3.75 (PS3), 5.0 (PS4), 8.0 (PS5 and PS6) and 11 ms (PS7-P8), respectively. The gradient strength was incremented in steps of 4 %, so that 23 points could be used for regression analysis. The recovery delay was set to 10 s. The number of scans per increment varied between 16 and 64 and typical experimental times were around 2 h to 6.5 h depending on the polymer concentration. All experiments were run without spinning. We performed three separate experiments for each polymer, varying the diffusion delay time (Δ) in increments of 25 ms to monitor and control convection effects, ensuring accurate and reliable diffusion measurements. The standard Bruker pulse program stebpgp1s1d, employing a stimulated echo sequence with bipolar pair pulses was utilized. Gradient recovery delays were always set to 1 ms. To check reproducibility, three different measurements with increasing

diffusion times were always carried out. All LMS diffusion processing was performed by using the DiffAtOnce software package available at https://www.diffatonce.com. [34] All of the data leading to the reported *D* values afforded lines whose correlation coefficients were above 0.999. TRAIn solutions were obtained by the use of the algorithm provided by Xu et al. [35] In the TRAIn algorithm, the number of iterations was controlled by applying a trust-region criterion, which helps ensure convergence by adaptively adjusting the step size based on the local curvature of the objective function. This criterion limits the search area during each iteration, preventing overly large or inefficient updates that could lead to divergence or slow convergence. Additionally, the sparsity-promoting l2 regularization was set to 1.05, which helps in promoting sparsity in the solution by penalizing larger values of diffusion coefficients. The choice of l2 regularization value fine-tunes the balance between fitting the diffusion coefficient and avoiding overfitting by ensuring that the diffusion values remain small and focused on the most relevant values.

2.3. Statistical analysis

To evaluate the accuracy of measurements obtained from NMR diffusion experiments, we performed a comprehensive statistical analysis to quantify errors and validate the methodology. Cross-validation was employed across a diverse range of solvents and concentrations, from 1.71 mg/mL to 150 mg/mL, to ensure that the method is broadly applicable and reliable under varying conditions. This approach provided robust error estimates, demonstrating high reliability across all tested algorithms. Most errors were below 0.5 %, with a maximum observed error of 3.5 %. At a 95 % confidence level, 95 out of 100 measurements are expected to exhibit errors below 3 %, with only 5 measurements exceeding this threshold.

The variability of diffusion values around the mean was quantified using the standard deviation (σ), and relative error was calculated by dividing the standard deviation by the mean (μ), providing a measure of accuracy. To visualize error distributions, the Probability Density

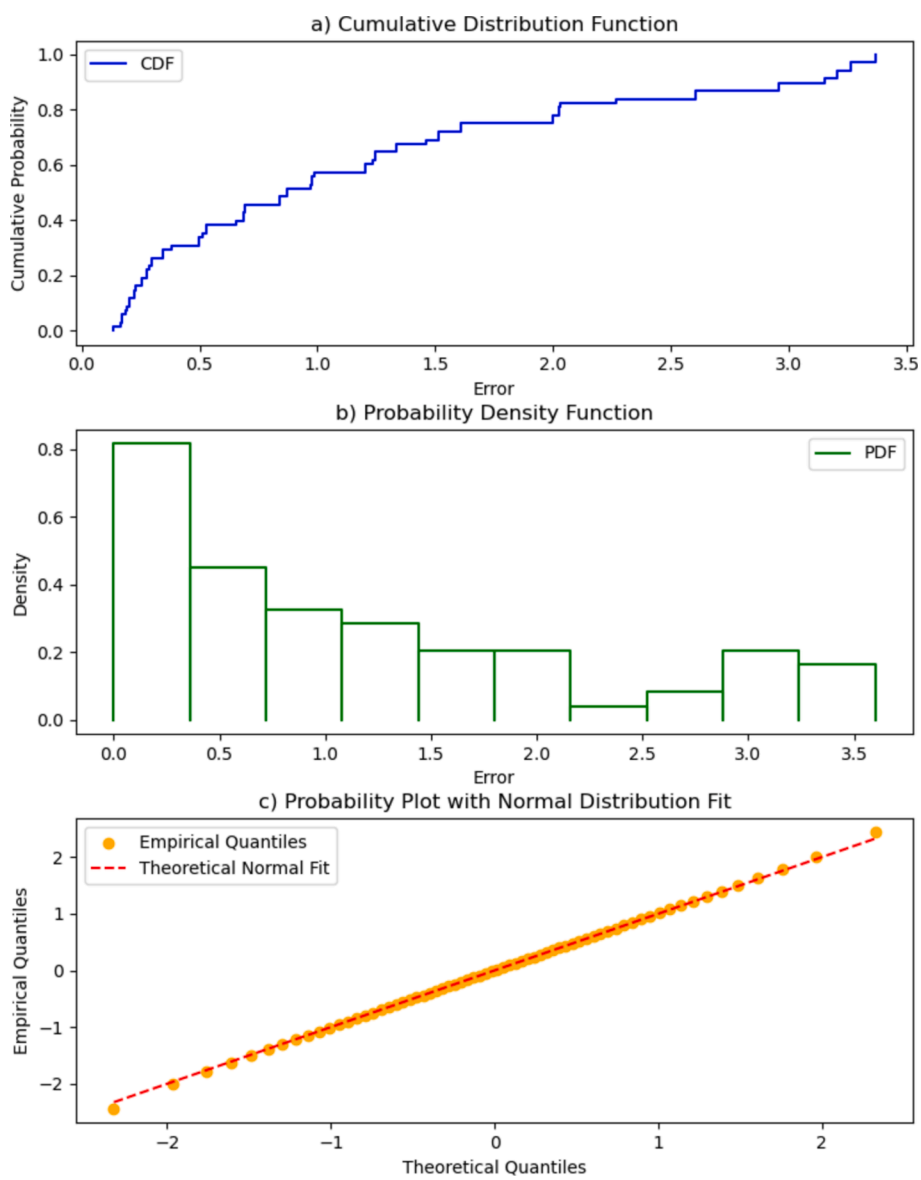


Fig. 1. a) Cumulative Distribution Function (CDF) showing the cumulative probability of errors in diffusion NMR measurements. The CDF illustrates the proportion of errors that fall below a given value, providing insight into the reliability of the method. b) Probability Density Function (PDF) highlighting the distribution of error magnitudes, emphasizing the most likely error values. c) Probability Plot comparing the empirical quantiles (observed errors sorted by magnitude) to theoretical quantiles derived from a normal distribution. The alignment of data points with the theoretical fit demonstrates that the majority of measurements closely follow a normal distribution, with 95% of errors falling below a 3% threshold, reinforcing the accuracy and consistency of the NMR diffusion methodology.

Function (PDF) was calculated, highlighting the most probable error values (Fig. 1a). The Cumulative Distribution Function (CDF), shown in Fig. 1b, illustrates the cumulative probability of errors less than or equal to a given value, offering a clear indication of overall reliability. Additionally, probability plots were generated (Fig. 1c) to evaluate the adherence of the empirical error distribution to theoretical models, such as the normal distribution, see the support information for details. This plot evaluates the alignment between the observed error distribution (empirical quantiles) and a theoretical normal distribution (theoretical quantiles).

The empirical quantiles were computed by sorting the observed error values in ascending order, representing the cumulative distribution of the experimental data. Theoretical quantiles were derived using the quantile function of the normal distribution, mapping evenly spaced to corresponding values on the theoretical normal scale. This approach enables a direct visual comparison, where deviations from the diagonal reference line indicate departures from normality. The close agreement observed in Figure c validates that the error distribution approximates a normal distribution, further supporting the reliability of the experimental measurements. These plots allowed for the identification of deviations, including those observed for PPG4 in THF. Despite these deviations, the overall validation confirms that the methodology is robust across varying conditions and exhibits reliable performance.

This analysis demonstrates that the model, derived from physical principles, accurately explains the observed experimental results and offers reliable predictions for diffusion coefficients and molecular weights under diverse solvent conditions and concentration ranges.

2.4. Method for molecular weight calculation

In this section, we describe the methodology of the iterative method used to determine the molecular weight of polymers based on their diffusion coefficients and solvent viscosity, independent of concentration and solvent type. This method is grounded on the previously described Eq. (9), Eqs. (11) and (12), allowing for molecular weight determination across a wide range of polymer concentrations and solvent types, circumventing the need for highly diluted samples traditionally required in traditional diffusion NNR technique.

The procedure is an iterative process which starts with an initial assumption for the molecular weight, typically using a low-value estimation of 2 Da. The $D\eta|_c$ is then calculated at a specific polymer concentration. Using Eqs. (11) and (12), the weight-average molecular weight Mw is iteratively refined by calculating new $D\eta|_c^*$ based on updated values for the molecular weight and comparing them with the experimentally measured values. If a significant difference between the calculated and measured values exists, the molecular weight is adjusted, and the process is repeated until convergence is achieved. This entire procedure is highly efficient, taking less than 200 ms per sample (see Fig. 3 below).

The iterative process was implemented using a combination of advanced computational tools. Matlab® scripts were developed to execute the routine, as provided in Code S1 and Code S2 for the two polymer systems tested. Diffusion coefficients were obtained using the DiffAtOnce (LMS) software, written in C#, and refined with the dART and TRAIn algorithms, both of which were coded in Matlab®. DOSY experiments were processed using the alternating direction sparse and low-rank unmixing (ADSpLRU) algorithm, also coded in Matlab®. These tools ensure robust diffusion coefficient estimarions, facilitating the high-precision calculations necessary for accurate molecular weight determination.

3. Results and Discussion

We propose a novel iterative methodology based on Eq. (9), Eqs. (11) and (12) to estimate molecular weight. This method begins with the measured value of $D\eta|_c$ at a specific concentration, starting with an assumed weight-average molecular weight (Mw) of 2 Da as a provisional solution (Scheme 1). Using this initial estimate of Mw, from Eq. (11) and (12) we estimate the value of $D\eta|_{1/\infty}$ and the scaling coefficient κ, respectively. Subsequently, with the help of Eq. (9) we derive a tentative value for $D\eta$ denoted as $D\eta|_c^*$. Recall that the value of $D\eta|_c$ is already known from the measured D value and the corresponding viscosity of the solvent used, and therefore the difference between the value of $D\eta|_c$ and the estimated $D\eta|_c^*$ is quantified as the methodological error. If the difference between $D\eta|_c$ and $D\eta|_c^*$ is high, the molecular weight increases a Δ value of 1 Da, then the method iterates once again returning a new estimate of $D\eta|_c^*$, that gets again matched with $D\eta|_c$, being the process consecutively repeated until the error is minimized (Eq. (14).

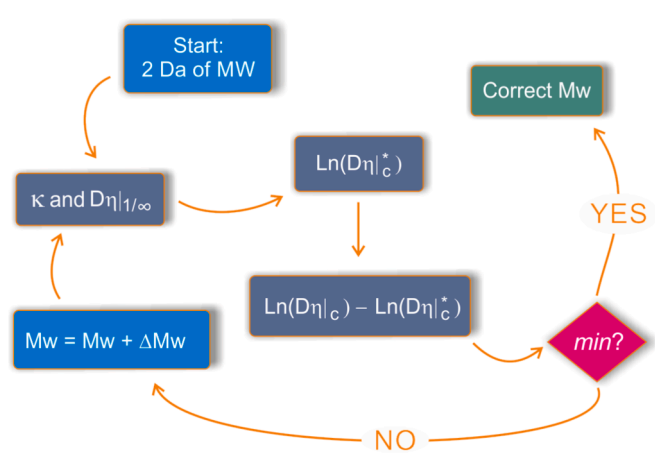
$$\min abs\left(D\eta|_c - D\eta|_c^*\right) \tag{14}$$

Scheme 1 provides the flow diagram detailing each step of the procedure. This iterative cycle refines the molecular weight estimate in order to gradually reduce the margin of error. This iterative approach is independent of the type of solvent and of the concentration of the polymer under study (see below), and it $D\eta|_c^*$ is therefore a versatile tool for determining molecular weight.

To validate the method at any concentration and with any type of deuterated solvent, we choose two different polymeric systems: i) a set of 4 monodispersed polypropylene glycol (PPG) polymers of molecular weight in the range of 0.45 to 5 kDa, and ii) a second set of 8 monodispersed polystyrene (PS) polymers of a much broader molecular weight spectrum ranging from 0.37 to 543 kDa.

For each polymer, ranging from PPG1 to PPG4 and from PS1 to PS8, we processed the spectra derived from the 24 gradient values obtained using a pulse field gradient-stimulated echo sequence with bipolar pair pulses (PGF-STE-BPP). [1] The analysis was carried out using several advanced software tools and scripts: the DiffAtOnce package (LMS), [34] a Matlab® script that implements the Algebraic Reconstruction Technique for Diffusion NMR, [37] and the Matlab® script developed by Kaipin et al. for TRAIn (Trust-Region Regularisation for Inversion problems). [35] Additionally, for the processing of DOSY data, we employed both the recently introduced SILT-DOSY [38], methodology, leveraging their capabilities to enhance the quality and accuracy of our analyses. For each PPG or PS polymer, six different incremental concentrations in weight were prepared from high dilution of ca. 0.1 wt% – 1.71 mg/mL – to 10 wt% – 150 mg/mL.

For each of the two polymers type we built a calibration curve based on ln(Dη) versus concentration. In overall, a set of 4 calibration curves for PPG and another set of 8 calibration curves for PS, all of them con-



Scheme 1. The iterative process for determining the molecular weight of the polymer with no dependence on solvent and concentration.

sisting of 6 points at different concentration, were obtained. All these plots are shown in Figs. S3-S14. From the slope of all these curves we extracted the Kappa value and from the independent term we obtained the $D\eta|_{1/\infty}$ value. With the estimated Kappa values, we built a new calibration curve called the "universal Kappa curve" (with the help of Eq. (12), and with the obtained $D\eta|_{1/\infty}$ values, we calculated a second curve called "universal infinite dilution curve" (based on Eq. (11). With these two curves in our hands, we were ready to apply the method described in Scheme 1. Fig. 2a–b illustrates the Kappa curve and the infinite dilution curve for the PS system. Figs. S15-S16 show the equivalent curves for the PPG system.

Interestingly, in the polystyrene (PS) system, two linear regimes were observed in the Kappa curve, with a discernible breakpoint at around 118 kDa. This behavior is likely to reflect a transition in polymer dynamics, characterized by an altered balance between excluded volume interactions and mean-field effects, such as chain entanglement and hydrodynamic interactions, which become more pronounced at higher molecular weights. The absence of a similar curvature in the Kappa curve for polypropylene glycol (PPG) polymers is probably due to the lower molecular weight range explored in this system, where excluded volume interactions dominate, and chain entanglements are less significant. This transition highlights the sensitivity of the method to changes in polymer dynamics and provides a unique opportunity to study these effects across different molecular weight regimes.

The application of LMS methods (via the DiffAtOnce® package), TRAIn and dART algorithms to the NMR signal attenuations allowed to retrieve the involved D-values. The obtained D values are almost the same, but the profile of both TRAIn and dART solutions slightly differ, i.e., TRAIn solutions are sharp with smooth sides whereas dART are broader but with edgy ends. Tables S1-S12 provide all the quantitative values for TRAIn solutions, whereas Tables S16-S31 provide all of them for LMS and dART solutions. It is important to mention that for all these experiments chloroform-d_1 was selected as the solvent of choice due to its intermediate viscosity ($\eta = 0.56714710^{-3} kgm^{-1}s^{-1}$ at 21 °C) and its widespread utilization. As described earlier in Scheme 1, each molecular weight is determined when the error between $D\eta|_c$ and the estimated $D\eta|_c^*$ is minimized by iteratively applying Eq. (9), Eqs. (11) and (12). Fig. 3 provides a clear visualization of how the algorithm minimizes the difference between the experimental diffusion coefficient and the predicted diffusion coefficient based on the molecular weight. The minimization metric used is $abs(D\eta|_c - D\eta|_c^*)/D\eta|_c$*100, represents the relative difference between the experimentally measured diffusion coefficient and the calculated diffusion coefficient during the iterative optimization process. This is not a direct experimental error but rather a convergence metric used by the algorithm to identify the molecular weight that best satisfies the scaling relationship.

The next step of our study was verifying the universality of our method in terms of the chosen deuterated solvent and demonstrate that independently of the solvent in where the polymer is soluble, the universal curves developed in chloroform are valid. To do that, we prepare several PS and PPG solutions not only at different concentrations (ranging from 0.1 wt% –. 1.71 mg/mL – to 10 wt% – 150 mg/mL) but also in different solvents such as benzene, toluene and THF. The polymers chosen were PPG3, PPG4, PS4 and PS6 with known molecular

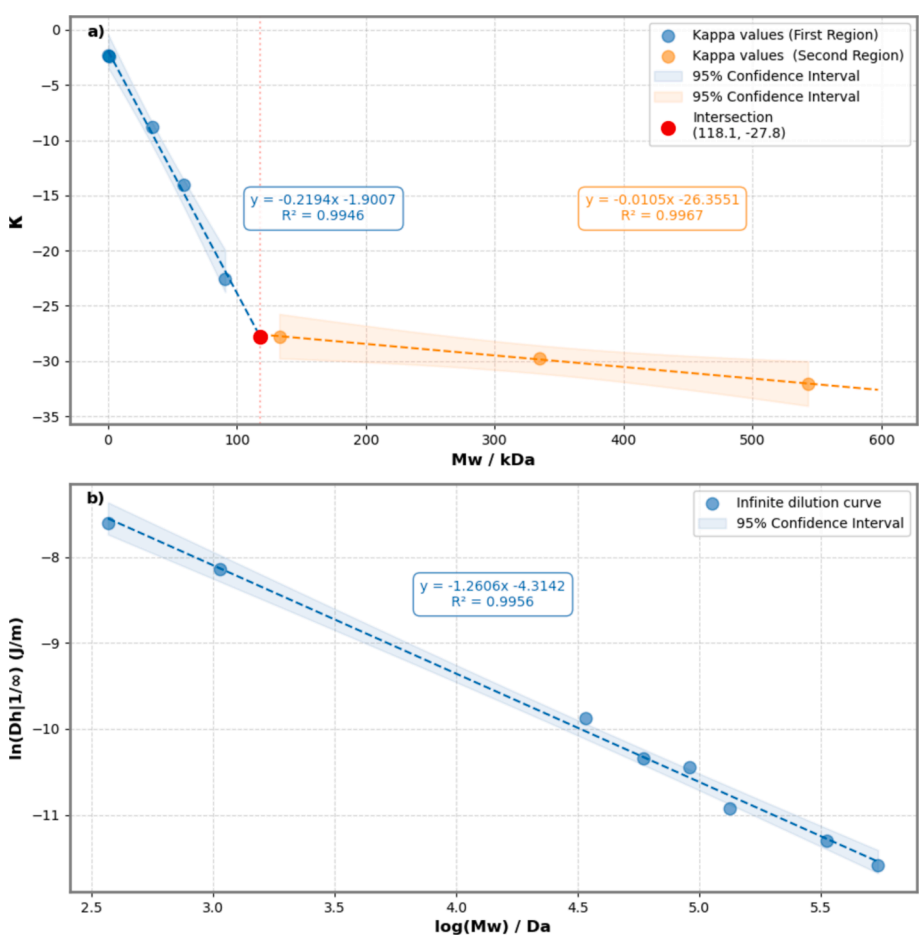


Fig. 2. A) Kappa curve and b) infinite dilution curve representing ln(Dη|_(1/∞)) vs log (Mw) for all the assayed PS polymers. All the D-values used in the construction of these curves were retrieved with the TRAIn algorithm. LMS and dART values can be found in Tables S16-S31. The confidence interval calculation is explained in the support information.

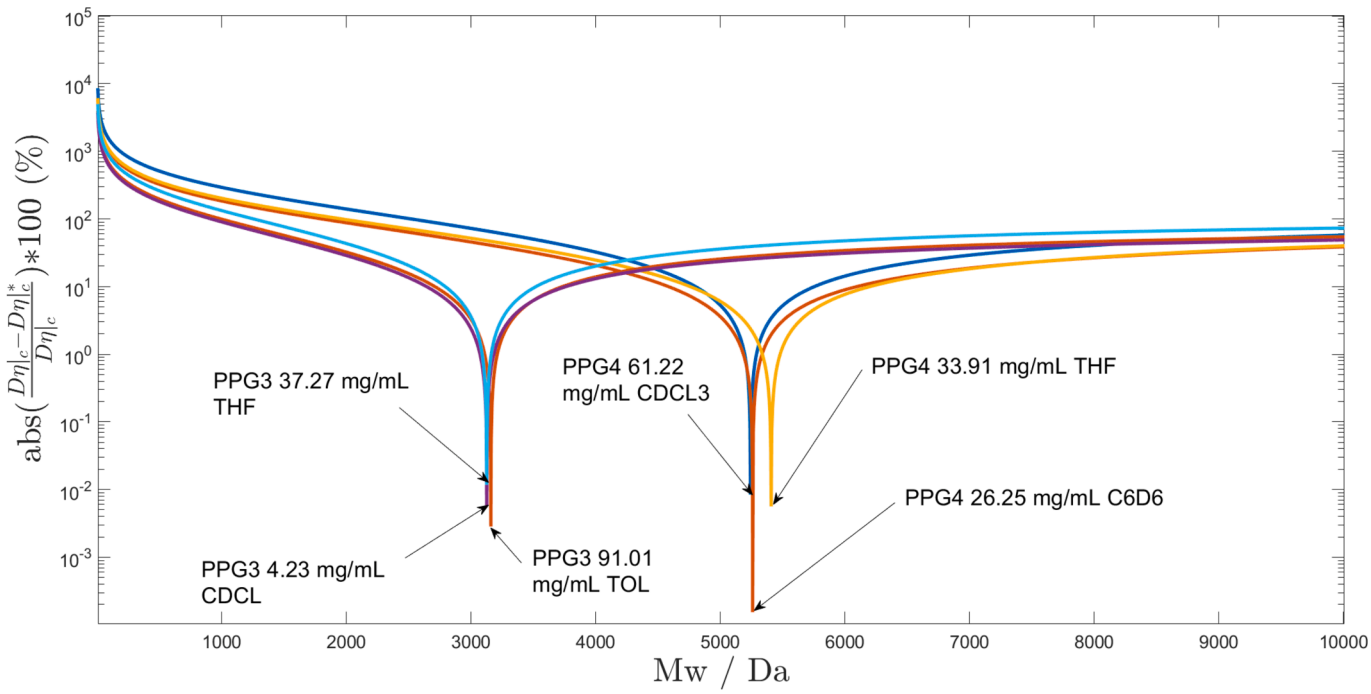


Fig. 3. Estimation of PPG molecular weight through the iterative process where the difference between $D\eta|_c$ and $D\eta|_c^*$ is minimized. The error is calculated as follows: $Error(\%) = abs(D\eta|_c - D\eta|_c^*)/D\eta|_c{*}100$. The Y-axis is in logarithmic scale.

weight values of 3250, 5300, 58,900 and 335000, respectively.

Table 2 shows the experimental Dη values, and the average molecular weight obtained using our described iterative method with four different polymers, each in three different solvents. In the case of PPG3, the calculated molecular weights were within the range of 3125 to 3159 Da, with the highest variability between the commercial and estimated value of 3.8 % found in the $CDCl_3$ series. Regarding PPG4, the obtained molecular weight window was 5242–5408 Da (Table 2), with THF-d_8 being the solvent in where the deviation of 2 % is the largest. The results for the two PS polymers assayed were also excellent. The molecular weight ranges were 58766–57048 Da and 343649–329058 Da for PS4 and PS7, respectively, with the highest error of 3.1 % found in THF for PS4 and 2.5 % in THF-d_8 for PS7.

Table 2
Diffusion values (m^2/s) [a] and predicted weight-average Mw for PPG3 and PPG4 in four different solvents and two different concentrations using the iterative process.

Solvent [b]	Polymer	C (mg/mL)	C (wt %)	Dη(10^{-3}) [b]	Av-Mw [c]	Diff [d]
Toluene-d_8	PPG3	91.01	10.5	0.0826	3126	3.7 %
THF-d_8	PPG3	37.27	3.3	0.1193	3159	2.8 %
$CDCl_3$	PPG3	4.23	0.5	0.1388	3125	3.8 %
Benzene-d_6	PPG4	26.25	3.0	0.0898	5259	0.8 %
THF-d_8	PPG4	33.91	3.9	0.0818	5408	2.0 %
$CDCl_3$	PPG4	61.22	6.5	0.0492	5242	1.1 %
Benzene-d_6	PS4	6.89	0.8	0.031550	58766	0.2 %
THF-d_8	PS4	12.35	1.3	0. 02726	57048	3.1 %
$CDCl_3$	PS4	71.49	4.8	0.016430	57616	2.2 %
Benzene-d_6	PS7	5.14	0.6	0. 01076	335570	0.1 %
THF-d_8	PS7	30.86	3.4	0. 00454	329058	1.7 %
$CDCl_3$	PS7	108.35	7.3	0. 00137	343649	2.5 %

[a] Experimental error in the D values is ± 3 % (see statistical analysis in the supporting information).
[b] The viscosity η used was taken from Perry's Chemical Engineers' Handbook 8th Edition (https://www.knovel.com) and is 0.656, 0.549, 0.491, 0.556 10^{-3} kg m^{-1} s^{-1} for benzene, toluene, THF and chloroform, respectively. For more details see Table S13.
[c] Weight-average molecular weight (Mw) in Dalton.
[d] The error is calculated in percentage as the difference between the estimated value and the value given by the commercial provider, divided by the value given by the commercial provider.

In all cases, the methodology described yielded average error rates below 4 %, underscoring its accuracy and consistency in determining molecular weights across various polymer types and solvent environments. This conservative estimation highlights the reliability of diffusion NMR as a complementary technique to SEC. The molecular weights obtained using this methodology closely align with those determined by SEC (Table 1), with deviations typically within 1–4 %. These differences can be attributed to the inherent variations in the principles underlying the two techniques: SEC relies on calibration standards and column separations, while diffusion NMR directly measures molecular motion in solution. SEC provides detailed molecular weight distributions and is ideal for routine analysis but requires calibration standards, consumables, and regular maintenance, making it less suitable for complex polymers or solvent-sensitive systems. Diffusion NMR, on the other hand, uses diffusion coefficients to infer molecular weight without external calibration and excels in solvent flexibility and complex polymer analysis. Although the dataset focuses on a subset of the polymers listed in Table 1, this selection represents a proof of concept for the diffusion NMR methodology. The chosen polymers span a wide range of molecular weights and were tested in multiple solvent environments and different concentrations to validate the applicability of the method under diverse conditions. Recent advancements, such as ultrafast (UF) and time-resolved (TR) diffusion NMR methods, have significantly reduced acquisition times, making the technique more competitive with SEC in terms of speed. Ongoing efforts are focused on adapting these methods for faster and more accurate polymer molecular weight predictions, potentially bridging the gap between the two techniques. Future work will expand the dataset to include additional polymers, further generalizing the approach and solidifying its role as a practical and efficient tool for polymer analysis.

4. Conclusions

An innovative iterative methodology for accurately estimating the

molecular weight of linear and narrow PPG and PS polymers has been described for the first time using both PGSE and SILT-DOSY NMR procedures. The main inputs to this method are the quantitative diffusion coefficient (D), solvent viscosity, and sample concentration. The method has been demonstrated to be valid in a wide range of concentrations, thanks to the developed Kappa and high dilution curves. These curves capture the growing polymeric interactions that occur as a function of concentration, helping to deduce true molecular weight at any concentration and in any deuterated solvent. The molecular weight predictions obtained by this method has achieved an absolute error below 4 % in all the cases, indicating its high accuracy and effectiveness. This new iterative method is presented in the Supporting Information as Code S1 and Code S2. Current research efforts are focused on exploring the broader application of this method to other polymers with higher polydispersity and especially those based on polysaccharides. Additionally, we are investigating the implications of the transitions observed in the Kappa plot, which likely reflect shifts in polymer dynamics due to excluded volume effects and mean-field interactions. These transitions hold potential for providing deeper insights into chain behavior, entanglements, and polymer–solvent interactions.

CRediT authorship contribution statement

Francisco Manuel Arrabal-Campos: Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Data curation. **Marta González Lázaro:** Investigation, Data curation. **Juana M. Pérez:** Investigation, Data curation. **Juan A. Martínez Lao:** Investigation, Data curation. **Ignacio Fernández:** Writing – review & editing, Supervision, Methodology, Funding acquisition, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research has been funded by the State Research Agency of the Spanish Ministry of Science and Innovation (PDC2021-121248-I00, PLEC2021-007774, PID2021-126445OB-I00 and CPP2022-009967) and by the Gobierno de España MCIN/AEI/10.13039/501100011033 and Unión Europea "Next Generation EU"/PRTR.

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.eurpolymj.2024.113710.

Data availability

Data will be made available on request.

References

[1] a) E.O. Stejskal, J.E. Tanner, Spin diffusion measurements: spin echoes in the presence of a time-dependent field gradient, J. Chem. Phys. 42 (1965) 288–292, https://doi.org/10.1063/1.1695690;
b) J.E. Tanner, Use of the stimulated echo in nmr diffusion studies, J. Chem. Phys. 52 (1970) 2523–2526, https://doi.org/10.1063/1.1673336;
c) C.S. Johnson, Diffusion ordered nuclear magnetic resonance spectroscopy: principles and applications, Prog. Nucl. Magn. Reson. Spectrosc. 34 (1999) 203–256, https://doi.org/10.1016/S0079-6565(99)00003-5.

[2] a) A. Macchioni, G. Ciancaleoni, C. Zuccaccia, D. Zuccaccia, Determining accurate molecular sizes in solution through NMR diffusion spectroscopy, Chem. Soc. Rev. 37 (2008) 479–489, https://doi.org/10.1039/B615067P;
b) L. Avram, Y. Cohen, Diffusion NMR of molecular cages and capsules, Chem. Soc. Rev. 44 (2015) 586–602, https://doi.org/10.1039/C4CS00197D;
c) P. Occhipinti, P.C. Griffiths, Quantifying diffusion in mucosal systems by pulsed-gradient spin-echo NMR, Adv. Drug Deliv. Rev. 60 (2008) 1570–1582, https://doi.org/10.1016/j.addr.2008.08.006;
d) R. Marega, V. Aroulmoji, F. Dinon, L. Vaccari, S. Giordani, A. Bianco, et al., Diffusion-ordered NMR spectroscopy in the structural characterization of functionalized carbon nanotubes, J. Am. Chem. Soc. 131 (2009) 9086–9093, https://doi.org/10.1021/ja902728w;
e) I. Åslund, D. Topgaard, Determination of the self-diffusion coefficient of intracellular water using PGSE NMR with variable gradient pulse length, J. Magn. Reson. 201 (2009) 250–254, https://doi.org/10.1016/j.jmr.2009.09.006;
f) G. Canzi, A.A. Mrse, C.P. Kubiak, Diffusion-ordered NMR spectroscopy as a reliable alternative to TEM for determining the size of gold nanoparticles in organic solutions, J. Phys. Chem. C 115 (2011) 7972–7978, https://doi.org/10.1021/jp2008557;
g) O. Kohlmann, W.E. Steinmetz, X.A. Mao, W.P. Wuelfing, A.C. Templeton, R. W. Murray, et al., NMR diffusion, relaxation, and spectroscopic studies of water soluble, monolayer-protected gold nanoclusters, J. Phys. Chem. B 105 (2001) 8801–8809, https://doi.org/10.1021/jp011123o.

[3] a) R. Marega, V. Aroulmoji, M. Bergamin, L. Feruglio, F. Dinon, A. Bianco, et al., Two-dimensional diffusion-ordered NMR spectroscopy as a tool for monitoring functionalized carbon nanotube purification and composition, ACS Nano 4 (2010) 2051–2058, https://doi.org/10.1021/nn100257h;
b) D. Šmejkalová, A. Piccolo, Host-guest interactions between 2,4-dichlorophenol and humic substances as evaluated by 1H NMR relaxation and diffusion ordered spectroscopy, Environ. Sci. Technol. 42 (2008) 8440–8445, https://doi.org/10.1021/es801809v;
c) D. Li, I. Keresztes, R. Hopson, P.G. Williard, Characterization of reactive intermediates by multinuclear diffusion-ordered NMR spectroscopy (DOSY), Acc. Chem. Res. 42 (2009) 270–280, https://doi.org/10.1021/ar800127e;
d) J.H. Lamm, P. Niermeier, A. Mix, J. Chmiel, B. Neumann, H.G. Stammler, et al., Mechanism of host-guest complex formation and identification of intermediates through NMR titration and diffusion NMR spectroscopy, Angew. Chemie – Int. Ed. 53 (2014) 7938–7942, https://doi.org/10.1002/anie.201402145;
e) R. Cao, A. Nonaka, F. Komura, T. Matsui, Application of diffusion ordered-1H-nuclear magnetic resonance spectroscopy to quantify sucrose in beverages, Food Chem. 171 (2015) 8–12, https://doi.org/10.1016/j.foodchem.2014.08.105;
f) W. Ge, J.H. Zhang, C.M. Pedersen, T. Zhao, F. Yue, C. Chen, et al., DOSY NMR: a versatile analytical chromatographic tool for lignocellulosic biomass conversion, ACS Sustain. Chem. Eng. 4 (2016) 1193–1200, https://doi.org/10.1021/acssuschemeng.5b01259;
g) Y.B. Monakhova, B.W.K. Diehl, T.X. Do, M. Schulze, S. Witzleben, Novel method for the determination of average molecular weight of natural polymers based on 2D DOSY NMR and chemometrics: example of heparin, J. Pharm. Biomed. Anal. 149 (2018) 128–132, https://doi.org/10.1016/j.jpba.2017.11.016.

[4] a) O. Söderman, P. Stilbs, NMR studies of complex surfactant systems, Prog. Nucl. Magn. Reson Spectrosc. 26 (1994) 445–482, https://doi.org/10.1016/0079-6565(94)80013-8;
b) W.S. Price, H. Ide, Y. Arata, Solution dynamics in aqueous monohydric alcohol systems, J. Phys. Chem. A 107 (2003) 4784–4789, https://doi.org/10.1021/jp027257z;
c) W.S. Price, F. Tsuchiya, Y. Arata, Time dependence of aggregation in crystallizing lysozyme solutions probed using NMR self-diffusion measurements, Biophys. J. 80 (2001) 1585–1590, https://doi.org/10.1016/S0006-3495(01)76131-0;
d) A. Gierer, K. Wirtz, Molekulare Theorie der Mikroreibung Zeitschrift Für, Naturforsch A 8 (1953) 532–538, https://doi.org/10.1515/zna-1953-0903;
e) H.C. Chen, S.H. Chen, Diffusion of crown ethers in alcohols, J. Phys. Chem. 88 (1984) 5118–5121, https://doi.org/10.1021/j150665a063;
f) F. Perrin, Mouvement brownien d'un ellipsoide - I. dispersion diélectrique pour des molécules ellipsoidales, J. Phys. Le Radium. 5 (1934) 497–511, https://doi.org/10.1051/jphysrad:01934005010049700;
g) F. Perrin, Mouvement Brownien d'un ellipsoide (II). Rotation libre et dépolarisation des fluorescences. Translation et diffusion de molécules ellipsoidales, J. Phys. Le Radium. 7 (1936) 1–11, https://doi.org/10.1051/jphysrad:01936007010100.

[5] a) K. Hayamizu, W.S. Price, A new type of sample tube for reducing convection effects in PGSE-NMR measurements of self-diffusion coefficients of liquid samples, J. Magn. Reson. 167 (2004) 328–333, https://doi.org/10.1016/j.jmr.2004.01.006;
b) F.M. Arrabal-Campos, L.M. Aguilera-Sáez, I. Fernández, A diffusion NMR method for the prediction of the weight-average molecular weight of globular proteins in aqueous media of different viscosities, Anal. Methods 11 (2019) 142–147, https://doi.org/10.1039/c8ay01817k;
c) J. Stepišnik, G. Lahajnar, I. Zupančič, A. Mohorič, Study of translational dynamics in molten polymer by variation of gradient pulse-width of PGSE, J. Magn. Reson. 236 (2013) 41–46, https://doi.org/10.1016/j.jmr.2013.08.003;
d) P. Stilbs, Fourier transform pulsed-gradient spin-echo studies of molecular diffusion, Prog. Nucl. Magn. Reson. Spectrosc. 19 (1987) 1–45, https://doi.org/10.1016/0079-6565(87)80007-9;
e) G. Kaysan, T. Rudszuck, L. Trapp, R. Balbierer, M. Kind, G. Guthausen, in: Recent applications of NMR diffusion experiments, 1st ed., Elsevier Ltd., 2022 https://doi.org/10.1016/bs.arnmr.2022.07.003.

[6] a) P. Groves, Diffusion ordered spectroscopy (DOSY) as applied to polymers, Polym. Chem. 8 (2017) 6700–6708, https://doi.org/10.1039/c7py01577a;
b) P. Groves, Diffusion ordered spectroscopy (DOSY) as applied to polymers, Polym. Chem. 8 (2017) 6700–6708, https://doi.org/10.1039/c7py01577a;
c) T.F. Nelson, C.P. Ward, Diffusion-ordered spectroscopy for rapid and facile determination of consumer plastic molecular weight, Anal. Chem. 95 (2023)

8560–8568, https://doi.org/10.1021/acs.analchem.3c00655;
d) N. Cherifi, A. Khoukh, A. Benaboura, L. Billon, Diffusion-ordered spectroscopy NMR DOSY: An all-in-one tool to simultaneously follow side reactions, livingness and molar masses of polymethylmethacrylate by nitroxide mediated polymerization, Polym. Chem. 7 (2016) 5249–5257, https://doi.org/10.1039/c6py00927a;
e) X. Guo, E. Laryea, M. Wilhelm, B. Luy, H. Nirschl, G. Guthausen, Diffusion in polymer solutions: molecular weight distribution by PFG-NMR and relation to SEC macromol, Chem. Phys. 218 (2017) 1–10, https://doi.org/10.1002/macp.201600440.

[7] a) D. Li, G. Kagan, R. Hopson, P.G. Williard, Formula weight prediction by internal reference diffusion-ordered NMR spectroscopy (DOSY), J. Am. Chem. Soc. 131 (2009) 5627–5634, https://doi.org/10.1021/ja810154u;
b) C.A. Crutchfield, D.J. Harris, Molecular mass estimation by PFG NMR spectroscopy, J. Magn. Reson. 185 (2007) 179–182, https://doi.org/10.1016/j.jmr.2006.12.004;
c) G. Hamdoun, C. Bour, V. Gandon, J.-N. Dumez, Empirical estimation of the molecular weight of gold complexes in solution by pulsed-field gradient NMR, Organometallics 37 (2018) 4692–4698, https://doi.org/10.1021/acs.organomet.8b00709;
d) T. Kubo, T. Nose, Diffusion of single chains in polymer matrices as measured by pulsed-field-gradient NMR: crossover from zimm- to rouse-type diffusion, Polym. J. 24 (1992) 1351–1361, https://doi.org/10.1295/polymj.24.1351;
e) K. Gu, J. Onorato, S.S. Xiao, C.K. Luscombe, Y.L. Loo, Determination of the molecular weight of conjugated polymers with diffusion-ordered NMR Spectroscopy, Chem. Mater. 30 (2018) 570–576, https://doi.org/10.1021/acs.chemmater.7b05063.

[8] C. Su, R. Hopson, P.G. Williard, Characterization of cyclopentyllithium and cyclopentyllithium tetrahydrofuran complex, J. Am. Chem. Soc. 135 (2013) 12400–12406, https://doi.org/10.1021/ja4059102.

[9] A. Chen, D. Wu, C.S. Johnson, Determination of molecular weight distributions for polymers by diffusion-ordered NMR, J. Am. Chem. Soc. 117 (1995) 7965–7970, https://doi.org/10.1021/ja00135a015.

[10] W. Li, H. Chung, C. Daeffler, J.A. Johnson, R.H. Grubbs, Application of (1)H DOSY for facile measurement of polymer molecular weights, Macromolecules 45 (2012) 9595–9603, https://doi.org/10.1021/ma301666x.

[11] W. Hiller, B. Grabe, The universal calibration for structure- and solvent-independent molar mass determinations of polymers using diffusion-ordered spectroscopy, Anal. Chem. 95 (2023) 18174–18179, https://doi.org/10.1021/acs.analchem.3c03797.

[12] a) S. Viel, D. Capitani, L. Mannina, A. Segre, Diffusion-ordered NMR spectroscopy: a versatile tool for the molecular weight determination of uncharged polysaccharides, Biomacromolecules 4 (2003) 1843–1847, https://doi.org/10.1021/bm0342638;
b) C. Barrère, M. Mazarin, R. Giordanengo, T.N.T. Phan, A. Thévand, S. Viel, et al., Molecular weight determination of block copolymers by pulsed gradient spin echo NMR, Anal Chem 81 (2009) 8054–8060, https://doi.org/10.1021/ac9018654;
c) N.E. Kuz'mina, S.V. Moiseev, V.I. Krylov, V.A. Yashkir, V.A. Merkulov, Quantitative determination of the average molecular weights of dextrans by diffusion ordered NMR spectroscopy, J. Anal. Chem. 69 (2014) 953–959, https://doi.org/10.1134/S1061934814100086;
d) P. Lewinski, S. Sosnowski, S. Kazmierski, S. Penczek, L-Lactide polymerization studied by 1H NMR with diffusion-ordered spectroscopy (DOSY): A "one NMR Tube Experiment" providing data on monomer conversion, polymer structure Mn and Mw, Polym. Chem. 6 (2015) 4353–4357, https://doi.org/10.1039/c5py00455a.

[13] Ulrich RD. P. J. Flory. In: Ulrich RD, editor. Macromol. Sci. First Edit, Boston, MA: Springer US; 1978, p. 69–98. Doi: 10.1007/978-1-4684-2853-7_5.

[14] Gennes P De. Scaling Concepts in Polymer Physics. Cornell University Press. 1979: 324.

[15] S. Augé, P. Schmit, C.A. Crutchfield, M.T. Islam, D.J. Harris, E. Durand, et al., NMR measure of translational diffusion and fractal dimension. application to molecular mass measurement, J. Phys. Chem. B 113 (2009) 1914–1918, https://doi.org/10.1021/jp8094424.

[16] a) F.M. Arrabal-Campos, L.M. Aguilera-Sáez, I. Fernández, A diffusion NMR method for the prediction of the weight-average molecular weight of globular proteins in aqueous media of different viscosities, Anal. Methods 11 (2019) 142–147, https://doi.org/10.1039/c8ay01817k;
b) F.M. Arrabal-Campos, L.M. Aguilera-Sáez, I. Fernández, Algebraic reconstruction technique for diffusion NMR experiments. application to the molecular weight prediction of polymers, J. Phys. Chem. A 123 (2019) 943–950, https://doi.org/10.1021/acs.jpca.8b08584.

[17] G.L. Pollack, Atomic test of the Stokes-Einstein law: Diffusion and solubility of Xe, PhysRevA 23 (1981) 2660–2663, https://doi.org/10.1103/PhysRevA.23.2660.

[18] G.L. Pollack, R.P. Kennan, J.F. Himm, D.R. Stump, Diffusion of xenon in liquid alkanes: Temperature dependence measurements with a new method. Stokes-Einstein and hard sphere theories, J. Chem. Phys. 92 (1990) 625–630, https://doi.org/10.1063/1.458413.

[19] F.M. Arrabal-Campos, P. Oña-Burgos, I. Fernández, Molecular weight prediction with no dependence on solvent viscosity. A quantitative pulse field gradient diffusion NMR approach, Polym. Chem. 7 (2016) 4326–4329, https://doi.org/10.1039/c6py00691d.

[20] W.S. Price, Pulsed-field gradient nuclear magnetic resonance as a tool for studying translational diffusion: Part 1, Basic Theory. Concepts Magn Reson 9 (1997) 299–336, https://doi.org/10.1002/(SICI)1099-0534(1997)9:5<299::AID-CMR2>3.0.CO;2-U.

[21] E. Ruzicka, P. Pellechia, B.C. Benicewicz, Polymer Molecular Weights via DOSY NMR, Anal. Chem. 95 (2023) 7849–7854, https://doi.org/10.1021/acs.analchem.2c05531.

[22] O. Tooley, W. Pointer, R. Radmall, M. Hall, V. Beyer, K. Stakem, et al., MaDDOSY (Mass determination diffusion ordered spectroscopy) using an 80 MHz bench Top NMR for the rapid determination of polymer and macromolecular molecular weight, Macromol. Rapid Commun. 45 (2024) 1–5, https://doi.org/10.1002/marc.202300692.

[23] I.W.F. Silva, A. McKay, A. Sokolova, T. Junkers, Towards the universal use of DOSY as a molar mass characterization tool: temperature dependence investigations and a software tool to process diffusion coefficients, Polym. Chem. 15 (2024) 1303–1309, https://doi.org/10.1039/D3PY01172K.

[24] S.F. Edwards, The size of a polymer molecule in a strong solution, J. Phys. A Math Gen. 8 (1975) 1670–1680, https://doi.org/10.1088/0305-4470/8/10/019.

[25] P.G. De Gennes, Dynamics of entangled polymer solutions II. inclusion of hydrodynamic interactions, Macromolecules 9 (1976) 594–598, https://doi.org/10.1021/ma60052a012.

[26] D.W. Schaefer, J.F. Joanny, P. Pincus, Dynamics of semiflexible polymers in solution, Macromolecules 13 (1980) 1280–1289, https://doi.org/10.1021/ma60077a048.

[27] C.M. Kok, A. Rudin, Relationship between the hydrodynamic radius and the radius of gyration of a polymer in solution, Die Makromol Chemie, Rapid Commun. 2 (1981) 655–659, https://doi.org/10.1002/marc.1981.030021102.

[28] G.D.J. Phillies, G.S. Ullmann, K. Ullmann, T. Lin, Phenomenological scaling laws for "'semidilute"' macromolecule solutions from light scattering by optical probe particles, J. Chem. Phys. 82 (1985) 5242–5246, https://doi.org/10.1063/1.448969.

[29] G.D.J. Phillies, Universal scaling equation for self-diffusion by macromolecules in solution, Macromolecules 19 (1986) 2367–2376, https://doi.org/10.1021/ma00163a006.

[30] K. Ullmann, G.S. Ullmann, G.D. Phillies, Optical probe study of a nonentangling macromolecule solution—bovine serum albumin: Water, J Colloid Interf. Sci. 105 (1985) 315–324, https://doi.org/10.1016/0021-9797(85)90305-4.

[31] G.S. Ullmann, K. Ullmann, R.M. Lindner, G.D.J. Phillies, Probe diffusion of polystyrene latex spheres in poly(ethylene oxide)-water, J. Phys. Chem. 89 (1985) 692–700, https://doi.org/10.1021/j100250a028.

[32] G.D.J. Phillies, Dynamics of polymers in concentrated solutions: the universal scaling equation derived, Macromolecules 20 (3) (1987) 558–564, https://doi.org/10.1021/ma00169a015.

[33] H.J.V. Tyrrell, K.R. Harris, in: Diffusion in Liquids, Elsevier, 1984, https://doi.org/10.1016/C2013-0-01002-X.

[34] DiffAtOnce® is a registered program developed by I. Fernández and F. M. Arrabal-Campos in the University of Almería. 2013.

[35] K. Xu, S. Zhang, Trust-region algorithm for the inversion of molecular Diffusion NMR Data, Anal. Chem. 86 (2014) 592–599, https://doi.org/10.1021/ac402698h.

[36] P. Gao, P.E. Fagerness, Diffusion in HPMC Gels.I. determination of drug and water diffusivity by pulsed-field-gradient spin-echo NMR, Pharm. Res. 12 (1995) 955–964, https://doi.org/10.1023/A:1016293911499.

[37] F.M. Arrabal-Campos, L.M. Aguilera-Sáez, I. Fernández, Algebraic reconstruction technique for diffusion NMR experiments. application to the molecular weight prediction of polymers, J. Phys. Chem. A 123 (2019) 943–950, https://doi.org/10.1021/acs.jpca.8b08584.

[38] B. Yuan, Y. Ding, G.M. Kamal, L. Shao, Z. Zhou, B. Jiang, et al., Reconstructing diffusion ordered NMR spectroscopy by simultaneous inversion of Laplace transform, J. Magn. Reson. 278 (2017) 1–7, https://doi.org/10.1016/j.jmr.2017.03.004.